

Data Centers and Water Stress in Texas, U.S.:
A Spatial Analysis of Infrastructure, Drought, and Environmental
Vulnerability

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1. Introduction

1.1 Research Question and Purpose

The rapid expansion of cloud computing and artificial intelligence has fueled an unprecedented buildout of data center infrastructure across the United States, with Texas emerging as one of the nation's fastest-growing markets. Data centers are among the most water-intensive industrial facilities, consuming millions of gallons per day for evaporative cooling systems (Mytton, 2021). At the same time, Texas has experienced increasingly severe drought driven by climate variability, raising urgent questions about long-term water availability and the environmental footprint of digital infrastructure.

This paper investigates the following research question: Are data center locations spatially associated with local drought severity patterns across Texas census tracts? A secondary question asks whether any association persists after controlling for socioeconomic characteristics, including poverty, education, health insurance coverage, and population size.

1.2 Hypotheses

H1 (Environmental Proximity): Census tracts in closer proximity to data centers will experience higher drought severity due to increased water demand from cooling operations.

H2 (Socioeconomic Vulnerability): Tracts with higher poverty rates and lower socioeconomic capacity will exhibit higher drought severity, reflecting environmental justice dynamics.

H3 (Spatial Clustering): Drought severity will show significant positive spatial autocorrelation, with high-drought areas clustering geographically instead of distributing randomly across Texas.

1.3 Literature Context

The rapid growth of cloud computing and AI has made data centers one of the most resource-intensive forms of modern infrastructure. Masanet et al. (2020) estimate that global data centers consumed approximately 200 terawatt-hours of electricity in 2018, and water usage has grown proportionally as AI workloads continue to expand. A single hyperscale facility can require millions of gallons of water per day for cooling, making geographic siting decisions increasingly important for regional water systems, especially in drought-prone states like Texas. Beyond resource consumption, scholars have raised questions about who bears the burden of this infrastructure. Drawing on a long tradition of environmental justice research, Pastor et al. (2001) documented that industrial facilities are disproportionately sited in lower-income and politically marginalized communities, a pattern that Pasek et al. (2023) argue now extends to data centers, which they describe as a new form of infrastructure burden for surrounding communities.

The spatial dimensions of the Texas drought are well-established in the literature. Nielsen-Gammon (2012) characterized the 2011 Texas drought as one of the most severe on record and highlighted the increasing frequency of multi-year drought events driven by large-scale climate systems. Hao and AghaKouchak (2013) further demonstrate that compound drought events, simultaneous extremes across precipitation and temperature, are increasing in frequency across the southern United States, compounding the long-run water stress facing Texas communities. These regional patterns indicate strong geographic clustering that persists across time and space. Understanding this spatial structure is important as Anselin (1988) demonstrated that ignoring spatial autocorrelation in regression analysis produces biased and inefficient estimates, directly affecting the spatial regression framework. Together, these studies establish

the policy urgency and the methodological foundation for analyzing the relationship between data center infrastructure and water stress in Texas.

2. Data and Methodology

2.1 Study Area and Unit of Analysis

The study area is the state of Texas. The geographic unit of analysis is the census tract, as defined by the 2020 U.S. Census delineation. Census tracts were chosen because they provide fine-grained spatial resolution to capture within-county variation in both environmental conditions and socioeconomic characteristics. After removing tracts with missing values on key variables, the analytical dataset includes 5,935 census tracts. Boundaries were obtained from the U.S. Census Bureau TIGER/Line shapefiles.

2.2 Data Sources

Drought Severity (Dependent Variable): Drought data were obtained from the U.S. Drought Monitor (USDM), which provides weekly polygon-based maps classified into five categories: D0 (abnormally dry) through D4 (exceptional drought), coded numerically as 0 – 4. The April 2026 drought shapefile was used. The polygon layer was spatially intersected with census tract boundaries in QGIS and a mean drought severity score (DM_mean) was calculated per tract using the Join Attributes by Location (Summary) tool.

Data Center Locations (Key Independent Variable): A dataset of 296 data center facilities in Texas was manually compiled from DataCenterMap, company investor disclosures, and publicly available addresses. Facilities include hyperscale campuses (Google, Meta, Microsoft, Amazon), colocation providers (Equinix, Digital Realty, CyrusOne), and cryptocurrency mining facilities (Riot Blockchain, IREN). Coordinates were geocoded via Google Maps and verified against

satellite imagery. This manual approach has limitations since unlisted or smaller facilities may be missing.

Distance to Nearest Data Center: For each census tract, distance to the nearest data center was calculated using the Distance to Nearest Hub algorithm in QGIS. Tract centroids were generated and distances computed from each centroid to the nearest data center point, in kilometers (after reprojection to EPSG:3857).

Socioeconomic Variables: Five tract-level controls were drawn from the ACS 5-Year Estimates (2024): poverty rate (S1701), uninsured rate (S2701), share of adults with bachelor's degree or higher (S1501), total population in thousands (B01003), and median household income (B19013). CSV files were joined to tract shapefiles using GEOID as the common key.

2.3 Data Cleaning and Preprocessing

All processing was conducted in QGIS 3.44.7. ACS files contained non-numeric entries (dashes for suppressed values; '250,000+' for top-coded income). These were fixed with conditional Field Calculator expressions, converting dashes to NULL and '250,000+' to 250000. Layers were reprojected from EPSG:4269/4326 to EPSG:3857 before distance calculations to make sure that the metric is accurate. Tracts with NULL on any modeled variable were excluded. Median household income was retained in the descriptive statistics but excluded from regression models due to missing data (N = 611 of 6,896 tracts).

2.4 GIS and Statistical Methods

The analysis proceeded in four stages: (1) choropleth mapping in QGIS to visualize spatial distributions; (2) Global Moran's I and LISA analysis in GeoDa 1.22 using a queen contiguity spatial weights matrix with 999 permutations; (3) OLS regression of drought severity on distance

to data centers and five socioeconomic controls; and (4) Spatial Lag (SLM) and Spatial Error (SEM) models to account for spatial dependence detected in OLS residuals. Model fit was compared using the Akaike Information Criterion (AIC). Software used: QGIS 3.44.7; GeoDa 1.22.

3. Results and Analysis

3.1 Descriptive Statistics

Table 1: Descriptive Statistics for Key Variables (Texas Census Tracts, N = 5,935)

Variable	N	Mean	Std Dev	Min	Max
Drought severity (DM_mean)	6,632	1.64	1.04	0.00	3.75
Distance to nearest data center (km)	6,786	21.95	27.75	0.06	185.42
Poverty rate (%)	6,834	15.06	11.76	0.00	100.00
Uninsured rate (%)	6,840	17.81	11.03	0.00	73.50
Education — bachelor’s+ (%)	6,848	31.82	20.90	0.00	97.40
Population (thousands)	6,896	4.38	2.37	0.00	44.36
Median household income (\$) †	611	78,558	36,279	13,359	246,500

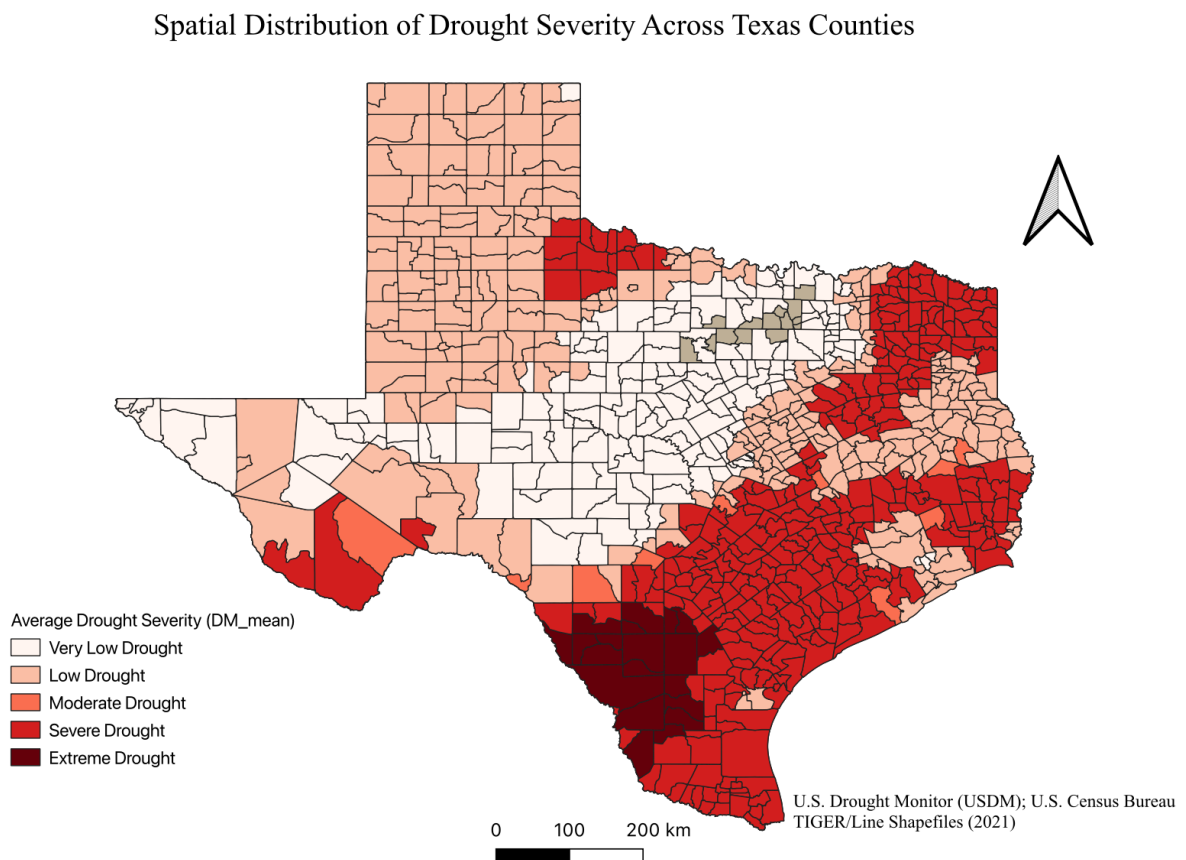
Note: All variables from ACS 5-Year Estimates 2024 and U.S. Drought Monitor (2026). † Median household income has substantial missing data (N = 611) due to ACS suppression of estimates for small tract populations; it is excluded from regression models.

Table 1 summarizes the distribution of all key variables. Mean drought severity is 1.64 (SD = 1.04), indicating that the average census tract falls between D0 (abnormally dry) and D1 (moderate drought) during the study period. The mean distance to the nearest data center is 21.95 km (SD=27.75 km), reflecting the strong geographic concentration of facilities in the Dallas–Fort Worth, Austin, and Houston metropolitan areas, with some tracts as far as 185 km from the nearest facility. Poverty averages 15.06% and the uninsured rate 17.81%, well above national averages, reflecting Texas’s status as the state with the highest uninsured population in the

country. Educational attainment (bachelor's degree or higher) averages 31.82%, with wide variation across tracts (SD = 20.90%).

3.2 Choropleth Maps

Figure 1: Drought Severity by Census Tract (Texas, April 2026)

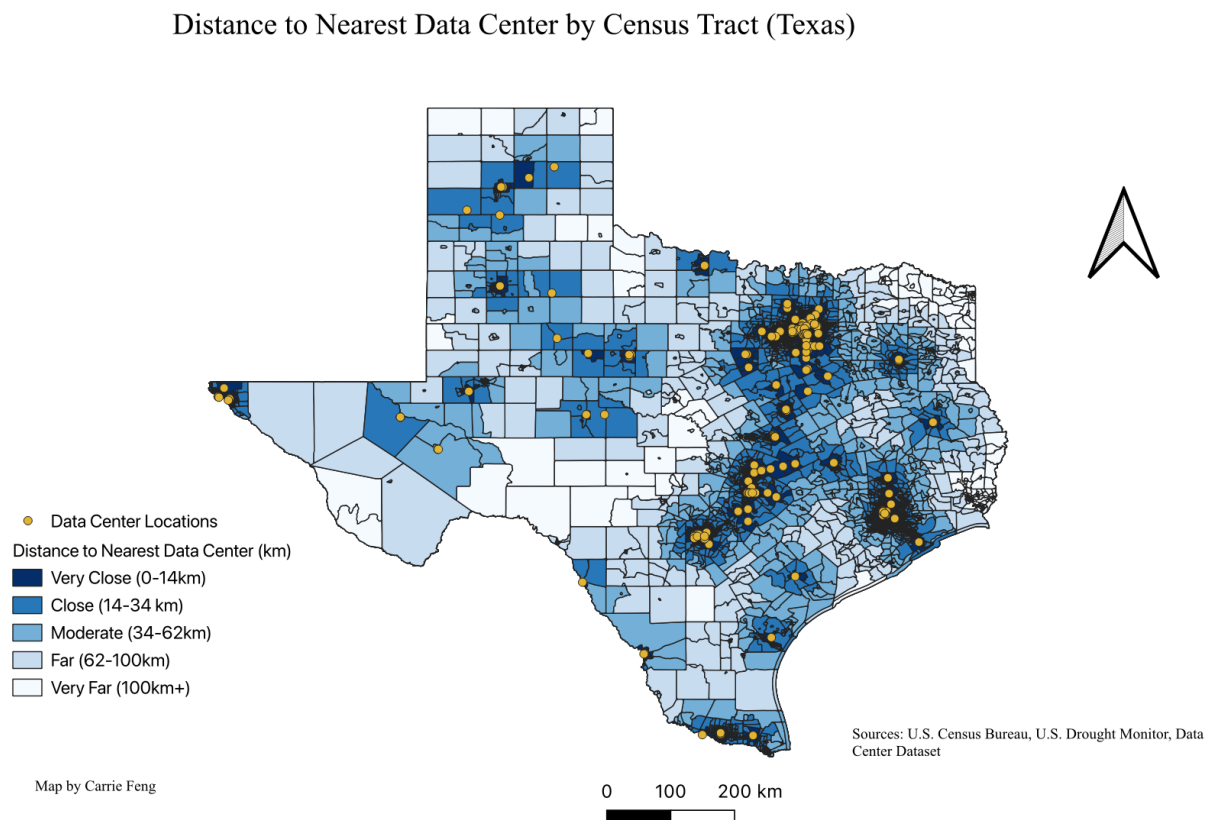


Classification method: Natural Breaks (Jenks), 5 classes.

Figure 1 reveals an interesting spatial pattern in drought severity across Texas. Contrary to expectations, the most severe conditions (Severe and Extreme Drought) are concentrated in South Texas and East Texas, particularly the Rio Grande Valley along the Mexico border and the eastern forested regions, as well as isolated pockets in the Panhandle. Central and West Texas, by contrast, show comparatively low drought severity during the April 2026 study period, appearing

in the Very Low to Low Drought categories. This east-south concentration of drought severity likely reflects the particular precipitation deficit patterns of the April 2026 snapshot instead of long-run climatic averages, as West Texas is typically the most chronically arid region of the state. The uneven geographic distribution nonetheless confirms that drought is not randomly distributed across census tracts, motivating the spatial autocorrelation analyses below.

Figure 2: Distance to Nearest Data Center by Census Tract

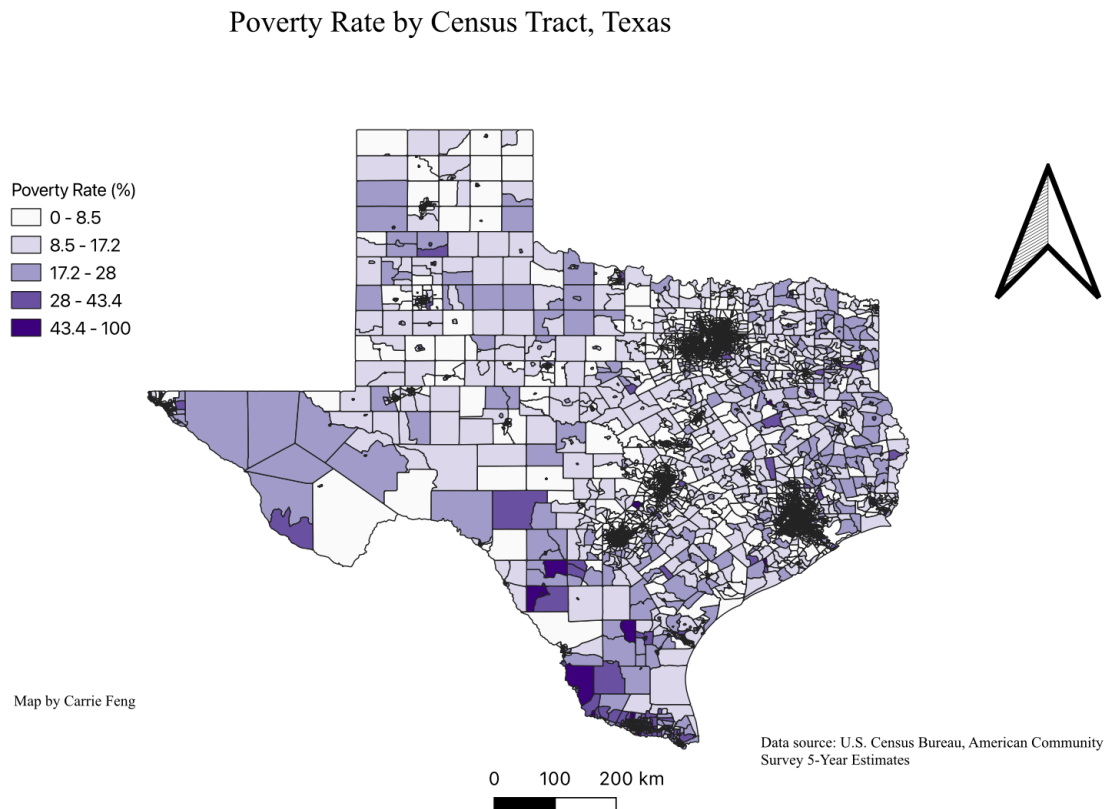


Classification method: Natural Breaks (Jenks), 5 classes.

Figure 2 illustrates the strong geographic concentration of data center infrastructure. Facilities cluster overwhelmingly in Dallas–Fort Worth, Austin, and Houston, areas with reliable power transmission and fiber connectivity. Western Texas, which is the most drought-affected region,

has almost no data center presence, with tracts located hundreds of kilometers from the nearest facility. This spatial juxtaposition between data center concentration and drought severity establishes the core empirical pattern motivating the regression analyses.

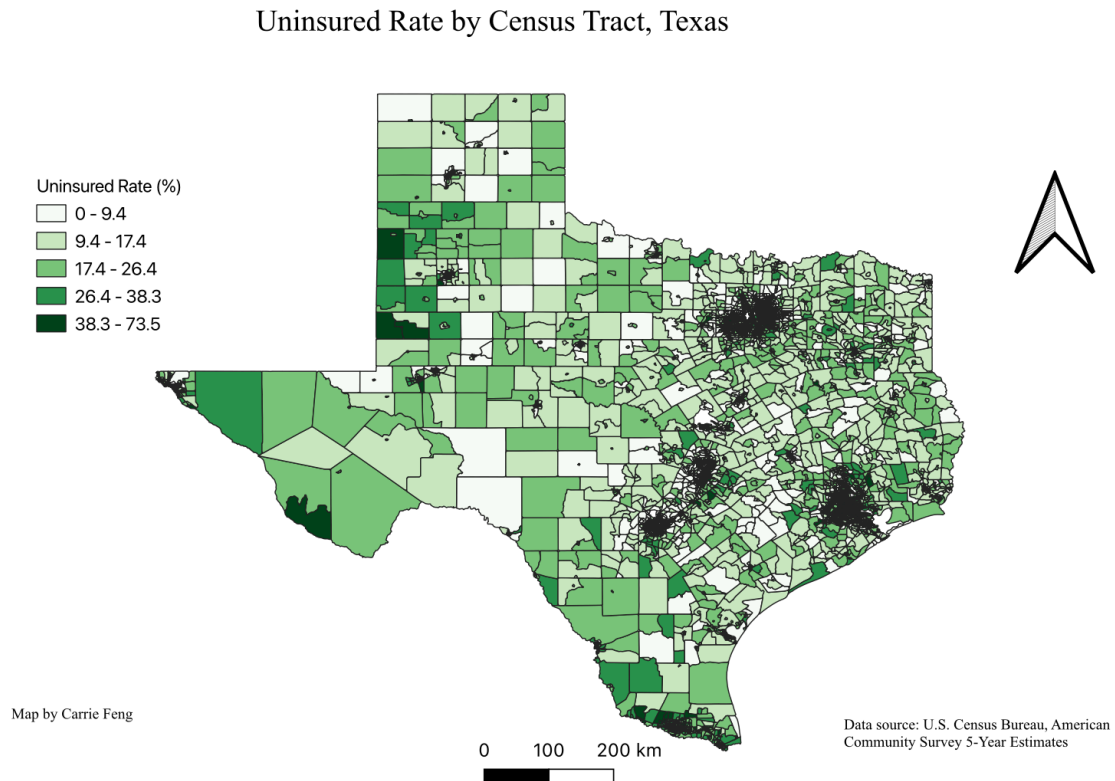
Figure 3: Poverty Rate by Census Tract, Texas (2024)



Classification method: Natural Breaks (Jenks), 5 classes.

Figure 3 shows poverty concentrated along the Texas-Mexico border corridor (Rio Grande Valley, El Paso), rural East Texas, and South Texas. These regions overlap substantially with high-drought areas from Figure 1, suggesting a spatial co-occurrence of socioeconomic vulnerability and environmental stress, a pattern consistent with the environmental justice literature (Pastor et al., 2001; Bullard, 1990).

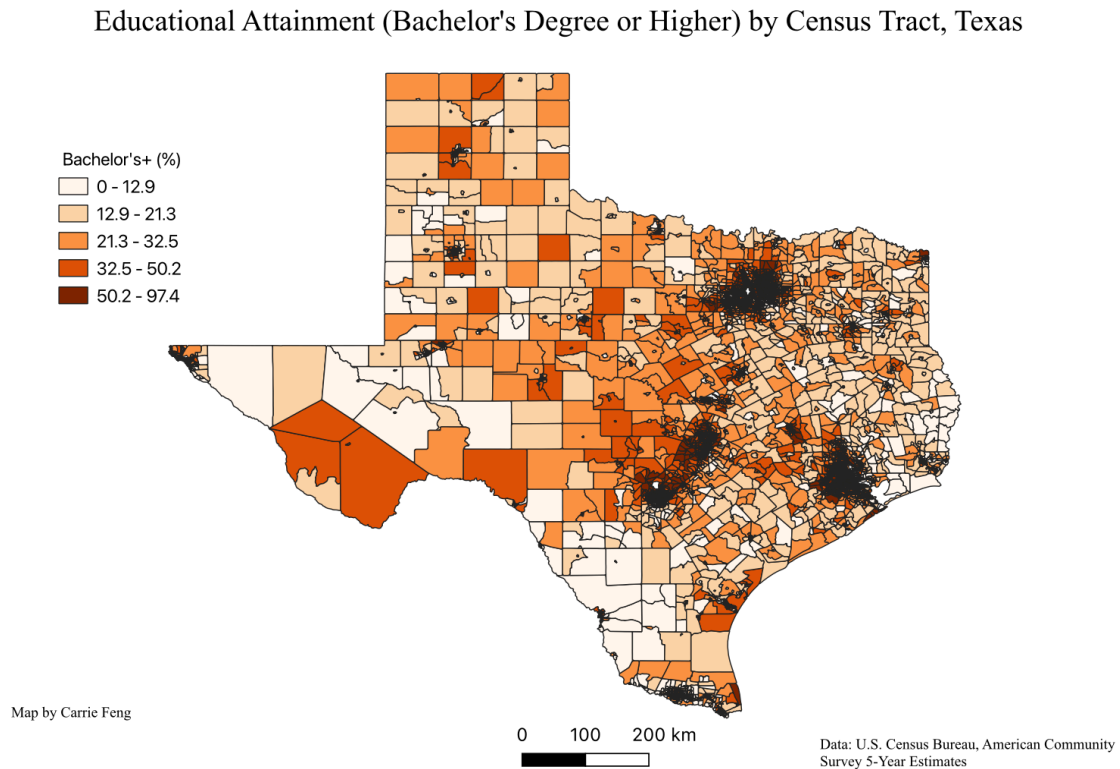
Figure 4: Uninsured Rate by Census Tract, Texas (2024)



Classification method: Natural Breaks (Jenks), 5 classes.

Figure 4 confirms Texas's status as the most uninsured state: the southern border, rural West and South Texas, and portions of Houston show the highest uninsured rates. The spatial pattern mirrors that of poverty (Figure 3) and drought (Figure 1), reinforcing the environmental justice framing and the theoretical basis for including this variable as a socioeconomic control.

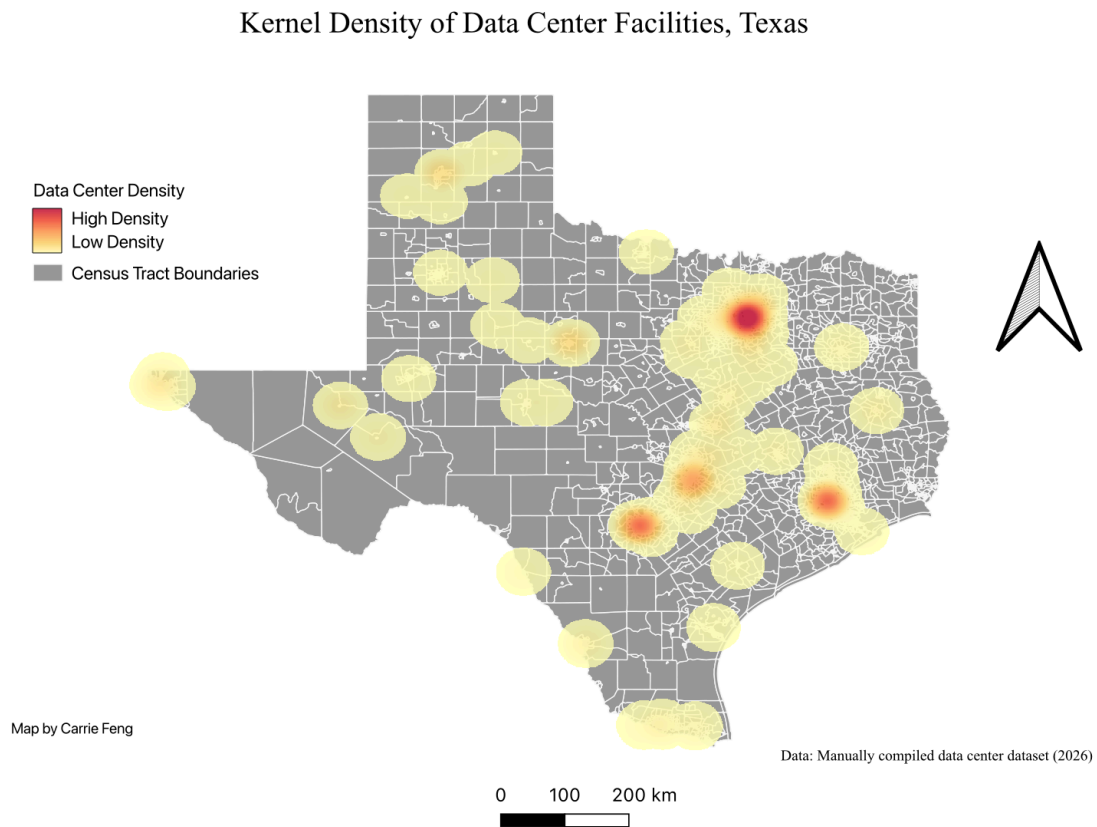
Figure 5: Educational Attainment (Bachelor's Degree or Higher) by Census Tract, Texas (2024)



Classification method: Natural Breaks (Jenks), 5 classes.

Figure 5 shows educational attainment concentrated in the major metropolitan areas, Austin, Dallas, Houston, with rural and border communities falling well below 10%. This pattern is nearly the inverse of the drought and poverty maps, confirming that educational capital is concentrated in the same lower-drought, higher-infrastructure regions where data centers also cluster.

Figure 6: Kernel Density of Data Center Locations

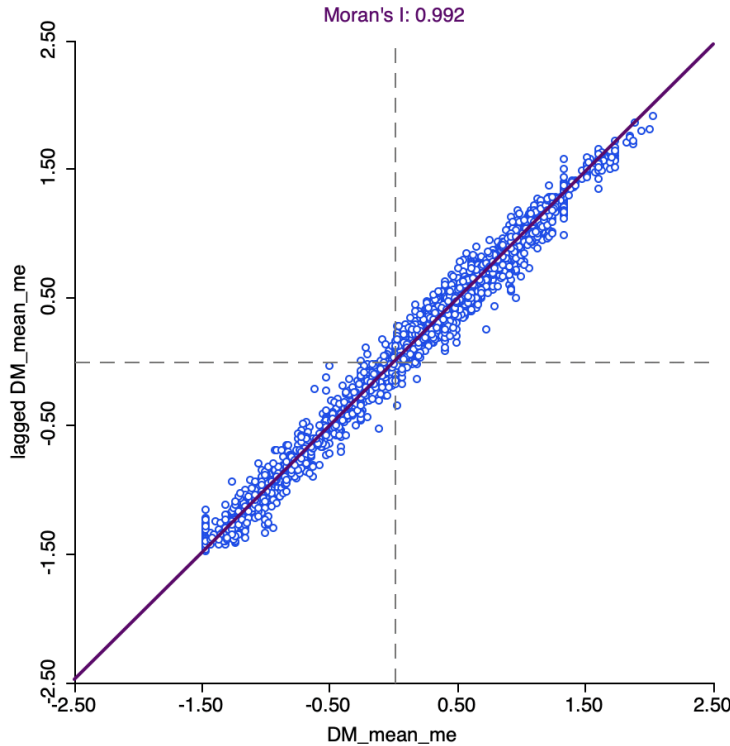


Classification method: Continuous, 5 classes.

Figure 6 presents a kernel density estimation of all 296 data center facilities in Texas, revealing a highly concentrated and geographically uneven distribution of digital infrastructure. The highest density hotspots, shown in deep red, are clustered in the Dallas–Fort Worth metroplex and the Austin–San Antonio corridor, with secondary concentrations around Houston and smaller clusters in Lubbock and the Panhandle. The regions with the greatest data center density correspond almost exactly with the areas of lowest drought severity identified in Figure 1, while the vast drought-stressed regions of South and West Texas show virtually no data center presence.

3.3 Global Spatial Autocorrelation

Figure 7: Moran's I Scatterplot — Drought Severity (DM_mean)

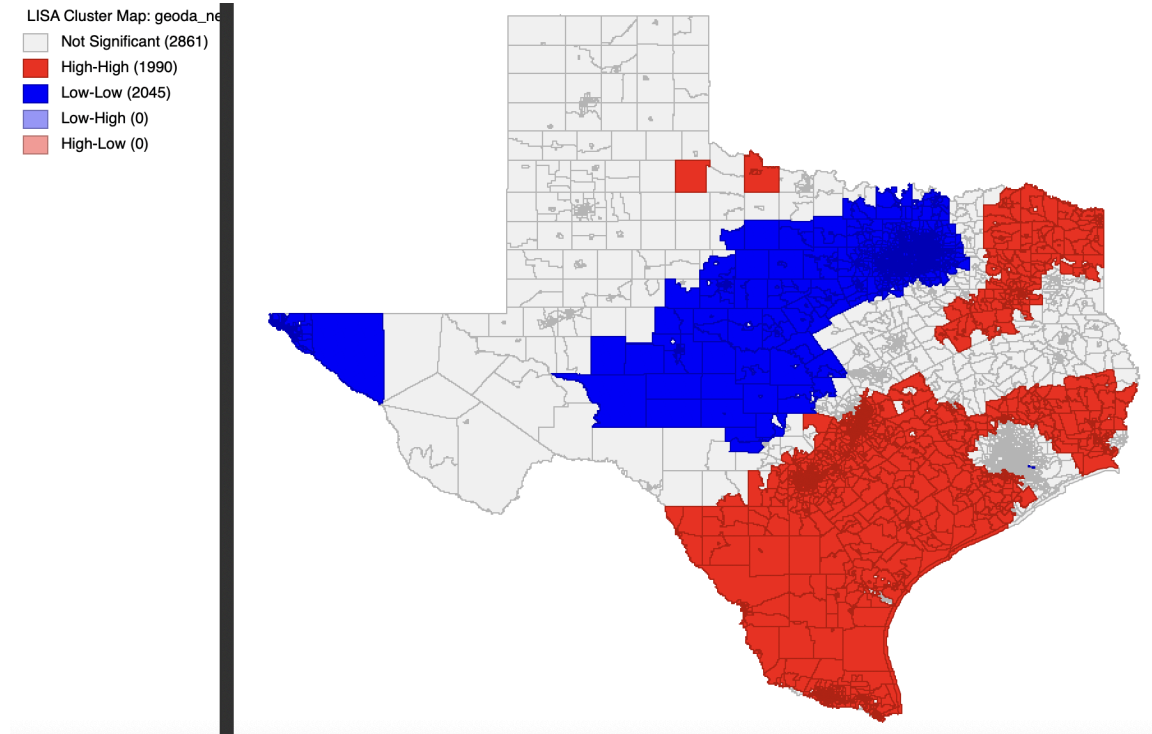


Source: GeoDa 1.22 spatial autocorrelation analysis; U.S. Drought Monitor (2026).

Global Moran's $I=0.992$ ($p=0.001$, 999 permutations), computed using a queen contiguity spatial weights matrix with $N=6,896$ observations, provides evidence of positive spatial autocorrelation. Tracts with high drought severity are systematically adjacent to other high-drought tracts, and low-drought tracts cluster together. The near-maximum I value indicates that drought in Texas is one of the most spatially structured phenomena in this dataset, consistent with the large-scale, regionally coherent nature of climate-driven drought documented by Nielsen-Gammon (2012) and Hao and AghaKouchak (2013). This result invalidates the OLS independence assumption, providing formal statistical justification for spatial regression models.

3.4 LISA Cluster Analysis

Figure 8: LISA Cluster Map — Drought Severity (Texas Census Tracts, N = 6,896)



Source: GeoDa 1.22 LISA analysis; U.S. Drought Monitor (2026); U.S. Census Bureau TIGER/Line Shapefiles.

The LISA cluster map (Figure 8) reveals statistically significant local clustering across 4,035 of 6,896 census tracts. High-High clusters, tracts with high drought surrounded by high-drought neighbors, are concentrated in southern and western Texas ($n=1,990$), forming a contiguous region of environmental stress from the Trans-Pecos through the Rio Grande Valley and South Texas. Low-Low clusters ($n=2,045$) appear primarily in northeast and central-east Texas, benefiting from Gulf moisture. The complete absence of High-Low and Low-High spatial outliers (both $n=0$) is consistent with the near-perfect Moran's I of 0.992, confirming that drought operates as a large-scale regional process instead of a combination of local conditions. This spatial structure provides an important baseline for interpreting the regression results below.

3.5 Regression Results

Table 2: OLS, Spatial Lag (SLM), and Spatial Error (SEM) Regression Results

Dependent Variable: Drought Severity (DM_mean) | N = 5,935

Variable	OLS β	OLS SE	SLM β	SLM SE	SEM β	SEM SE
Constant	2.143***	(0.134)	0.018	(0.020)	2.017***	(0.148)
Distance to data center (km)	0.008***	(0.001)	0.000	(0.001)	0.001	(0.001)
Poverty rate (%)	0.017***	(0.003)	0.000	(0.001)	0.002	(0.002)
Uninsured rate (%)	-0.013***	(0.003)	-0.000	(0.001)	-0.001	(0.002)
Education — bachelor's+ (%)	-0.003***	(0.001)	-0.000	(0.000)	-0.000	(0.001)
Population (thousands)	0.015**	(0.005)	0.001	(0.002)	0.002	(0.003)
ρ (spatial lag)	—	—	0.998***	(0.001)	—	—
λ (spatial error)	—	—	—	—	0.254***	(0.018)
R ²	0.080		0.999		0.210	
AIC	8,590.14		8,372.83		8,364.88	
N	5,935		5,935		5,935	

Note: Standard errors in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Spatial weights: queen contiguity. ρ = spatial autoregressive coefficient (SLM); λ = spatial error coefficient (SEM). SEM achieves lowest AIC and is the preferred model. SLM and SEM standard errors for socioeconomic predictors are approximate; verify against GeoDa output.

3.6 OLS Model

The OLS model explains approximately 8% of the variation in drought severity ($R^2=0.080$; F-statistic significant at $p < 0.001$). All five independent variables are statistically significant. Distance to data centers carries a positive coefficient ($\beta=0.008$, $p < 0.001$), indicating that tracts farther from data centers tend to have slightly higher drought severity. This result does not support H1 as stated; instead, it explains that geographic confounding, data centers cluster in urban areas with better infrastructure and lower drought risk, while the most drought-affected regions of the state are almost entirely without data center presence. Poverty shows a positive

relationship ($\beta=0.017$, $p < 0.001$), supporting H2: higher poverty is associated with more severe drought. The uninsured rate shows a negative coefficient ($\beta = -0.013$, $p < 0.001$), education a negative coefficient ($\beta = -0.003$, $p < 0.001$), and population a positive coefficient ($\beta = 0.015$, $p < 0.01$). Diagnostic tests reject normality (Jarque-Bera, $p < 0.001$) and detect heteroskedasticity (Breusch-Pagan, $p < 0.001$). Lagrange Multiplier tests for both spatial lag and spatial error dependence are highly significant ($p < 0.001$), confirming that spatial regression models are required.

3.7 Spatial Lag and Spatial Error Models

The Spatial Lag Model introduces a spatially lagged dependent variable ($W \cdot \text{Drought}$). The autoregressive coefficient Rho ($\rho=0.998$, $p < 0.001$) is very big and significant, indicating that each tract's drought severity is almost entirely predicted by its neighbors' values. Once this spatial structure is absorbed, all socioeconomic variables and the data center distance variable lose statistical significance. The AIC drops substantially from the OLS baseline (8,372.83 vs. 8,590.14), confirming improved fit.

The Spatial Error Model treats spatial dependence as operating through the error term. Lambda ($\lambda=0.254$, $p < 0.001$) is positive and significant, indicating that unobserved spatially structured factors, such as regional climate, soil type, and hydrology, drive drought in ways correlated across neighboring tracts. The SEM achieves the lowest AIC (8,364.88), making it the preferred model. As in the SLM, all predictors lose significance once spatial dependence is accounted for, reinforcing the conclusion that the OLS relationships were largely artifacts of geographic confounding instead of causal effects.

3.8 Hypothesis Testing

H1 — Not Supported as stated: The OLS coefficient on data center distance is positive (farther = more drought) rather than negative, and becomes insignificant in spatial models. Data centers are sited in lower-drought metropolitan areas; there is no evidence that they cause drought reduction.

H2 — Partially Supported: Poverty shows a significant positive relationship with drought in OLS, consistent with H2. However, this relationship disappears in spatial models, suggesting it reflects geographic co-occurrence of poverty and drought, and not an independent causal mechanism.

H3 — Strongly Supported: Moran's $I = 0.992$ ($p=0.001$) provides a lot of support. Drought is highly spatially clustered, with clear High-High ($n=1,990$) and Low-Low ($n=2,045$) regions confirmed by LISA analysis, and zero spatial outliers.

4. Discussion and Interpretation

4.1 Key Findings

Three core findings emerge from this analysis. First, drought severity in Texas is shaped by large-scale spatial processes. The Moran's I of 0.992, near the theoretical maximum, and the LISA cluster analysis confirm that drought is a regionally structured phenomenon driven by climate systems instead of local factors, a finding consistent with Nielsen-Gammon (2012) and Hao and AghaKouchak (2013). Second, data centers in Texas are concentrated in the state's major metropolitan areas (Dallas, Austin, Houston), which are also the areas with the lowest drought severity. This geographic pattern likely reflects rational infrastructure siting: facility

operators preferentially locate where power transmission, fiber connectivity, and water availability are most reliable. Third, once spatial dependence is accounted for through spatial regression, the apparent relationships between drought severity and both data center proximity and socioeconomic variables disappear entirely, suggesting that the OLS findings were driven by how poverty, low education, and distance from data centers all occur in the same drought-prone regions of Texas.

4.2 Environmental Justice Implications

Despite the limitations of the statistical models, the descriptive and mapping analysis reveals a meaningful spatial pattern with environmental justice significance. The areas of greatest drought severity in Texas are also the areas with the highest poverty rates, highest uninsured rates, and lowest educational attainment. These communities face a compounding set of vulnerabilities: chronic environmental stress, limited economic capacity for climate adaptation, and minimal proximity to the data center investment that has brought economic development and tax revenue to the Dallas–Austin–Houston triangle. Notably, as shown in Figure 6, data center infrastructure is almost entirely absent from the drought-stressed regions of South and East Texas, meaning these communities bear the environmental burden of statewide water stress without receiving the economic benefits of the digital infrastructure buildout. This spatial divergence is consistent with Bullard (1990) and Pastor et al. (2001), who documented the tendency for environmental burdens and economic development benefits to be distributed unequally across social and geographic lines.

While this study cannot establish a causal relationship between data center siting and drought burden, the geographic pattern raises important policy questions. As AI and cloud computing continue to drive data center demand, ensuring that this growth does not further strain water resources in already-vulnerable communities requires proactive regional planning and water rights oversight. Several Texas municipalities have already begun negotiating water use agreements with data center operators, suggesting that this concern is not hypothetical.

4.3 Spatial Dependence and Model Choice

A methodologically important contribution of this paper is its demonstration that OLS regression is not appropriate for analyzing drought outcomes at the census tract level. The Moran's I statistic of 0.992 on the dependent variable, combined with highly significant Lagrange Multiplier tests on OLS residuals, makes the case for spatial modeling essentially irrefutable. The spatial models confirm that the primary driver of tract-level drought is the regional drought environment, rather than any local characteristic measurable in administrative data. This finding should inform future research on environmental outcomes in Texas: any regression analysis of spatially patterned outcomes must account for spatial dependence, or risk severely overstating the explanatory power of local predictors.

4.4 Limitations

Data limitations: The data center dataset was manually compiled and may be incomplete, particularly for smaller or unlisted facilities. This introduces measurement error in the key independent variable and likely attenuates any true relationship between proximity and drought.

Additionally, a single monthly drought snapshot (April 2026) is used rather than a multi-year average, which may not represent long-run drought burden.

Methodological limitations: The analysis omits key environmental predictors, precipitation, temperature, elevation, and soil type, that would be expected to explain much of the spatial variation in drought. Future work should incorporate climate raster data from NOAA or PRISM. Robustness checks using county-level aggregation or alternative spatial weights specifications are also recommended. Finally, the cross-sectional design limits causal inference: panel designs incorporating data center construction dates would provide stronger evidence.

A further temporal consideration deserves emphasis: many of the data centers in the Texas dataset are either newly operational or still under construction. The water consumption footprint of large-scale data center facilities, especially hyperscale AI campuses, typically takes several years to fully materialize as facilities ramp up capacity and cooling demand. It is therefore plausible that any localized hydrological signal from data center water use is not yet detectable at the census tract level using current drought monitoring data. Future research should revisit this question using multi-year drought indices and longitudinal facility operational data as more time elapses and the cumulative water consumption of Texas's data center buildout becomes measurable.

5. Conclusion

This paper examined the spatial relationship between data center infrastructure and drought severity across 5,935 census tracts in Texas. The central finding is that drought in Texas is driven by large-scale spatial processes (Moran's $I = 0.992$), and that local variables, including proximity to data centers and socioeconomic characteristics, explain only modest variation in drought

outcomes at the tract scale. The Spatial Error Model, which achieved the best statistical fit (lowest AIC = 8,364.88), reveals that much of the apparent relationship between these variables and drought in OLS models is a product of geographic confounding: data centers cluster in lower-drought metropolitan areas not because they reduce drought risk, but because those areas offer the infrastructure and resource stability that operators require.

The descriptive analysis identifies a meaningful and policy-relevant spatial pattern: the communities experiencing the greatest drought burden in Texas are predominantly low-income, high-poverty communities along the southern border and in rural South and East Texas, the same communities that have seen the least data center investment and economic development. This spatial divergence reflects broader patterns of uneven infrastructure investment documented in the environmental justice literature (Bullard, 1990; Pastor et al., 2001). As Texas continues to attract data center investment driven by AI and cloud computing demand, proactive water resource planning and proper infrastructure investment will become increasingly critical to ensuring that the costs and benefits of the digital economy are distributed fairly.

Finally, this study should be understood as a first look at a rapidly evolving phenomenon. The data centers analyzed here range from long-established facilities to campuses that broke ground within the past two years, and the April 2026 drought snapshot captures a moment at which the full water consumption footprint of Texas's AI-driven infrastructure buildout has not yet had time to manifest in regional hydrological patterns. The absence of a detectable spatial signal between data center proximity and drought severity may reflect independence, or perhaps it is too early to tell. As someone deeply invested in this question, I intend to revisit this project in future research, incorporating multi-year drought trends, facility-level water use disclosures, and

panel data methods that can track how the relationship between digital infrastructure and water stress evolves over time.

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